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Perspective Article

Amyloid aggregation at solid-liquid interfaces: Perspectives of studies using model surfaces

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ABSTRACT

The aggregation of proteins and peptides into cytotoxic oligomeric and fibrillar amyloid aggregates represents an important mechanism in the development of numerous degenerative diseases such as Alzheimer's disease, Parkinson's disease, and type 2 diabetes mellitus. Amyloid aggregation is notoriously hard to control both *in vivo* and *in vitro*, because aggregation kinetics, aggregate morphology, and molecular aggregate structure are influenced by a multitude of environmental parameters, including monomer concentration, ionic strength, temperature, pH, and the presence of interfaces. While for a long time, amyloid aggregation has been studied mostly in bulk solution, surface and interface-related effects have recently become the focus of intense research efforts. Understanding the complex influences that solid-liquid interfaces exert on the aggregation of amyloidogenic proteins and peptides not only promises deeper insights into the molecular mechanisms of *in-vivo* membrane toxicity, but may also result in novel therapeutic approaches utilizing for instance amyloid-inhibiting nanoparticles. A particularly important route toward this goal is represented by studies using model surfaces with well-defined physicochemical properties in order to elucidate the involved physical and chemical mechanisms and interactions on a molecular level. This perspective article will provide an overview over the experimental techniques and model surfaces that so far have been utilized in such studies, summarize the obtained findings and insights, and address future challenges.

1. Introduction

To date, more than 40 diseases in humans have been associated with the aggregation of (partially) unfolded peptides and proteins into cytotoxic amyloid aggregates via the formation of intermolecular β sheets [1,2]. These so-called misfolding diseases comprise neurodegenerative diseases including Alzheimer's, Parkinson's, Huntington's, and prion disease, non-neuropathic systemic amyloidoses, and various non-neuropathic localized diseases such as cataract and type 2 diabetes mellitus. Several characteristics of amyloid formation hint at a nucleated growth mechanism, which typically is found to proceed in three distinct stages (see Fig. 1a) [3]. In an initial lag phase, thermodynamically disfavored oligomeric nuclei are assembled which then grow rapidly into larger, thermodynamically stable fibrillar aggregates in the so-called elongation phase. This growth proceeds via addition of monomers or smaller oligomers from solution, but may also involve secondary aggregation pathways such as fibril fragmentation and secondary nucleation [4,5]. The final plateau phase represents a saturation regime characterized by monomer depletion, in which a variety of morphologically different fibrillar structures may be found. These amyloid fibrils typically consist of rather thin protofibrils that associate with each other and may thereby form a large variety of different species of mature fibrils with various straight, coiled, or ribbon-like morphologies (see Fig. 1b) [3,6–9]. Although their morphology may differ substantially, virtually all amyloid fibrils are characterized by a cross-β structure with the β strands arranged perpendicular to the fibril axis [3]. Amyloid fibrils in general are cytotoxic with their toxicity correlating with fibril morphology [10–12]. Pre-fibrillar oligomeric structures, however, are currently considered the most toxic species [13], although recent studies suggest non-toxic oligomeric species may exist as well [14].

Amyloid aggregation is notoriously sensitive toward environmental conditions including ionic strength, pH, temperature, and concentration (see Fig. 2), small variations of which may have drastic effects on aggregation kinetics and amyloid morphology [6,9]. In the physiological environment, amyloid formation additionally occurs in the presence of various interfaces, most importantly cell membranes. While it has been established that the interaction with the various amyloid structures may result in membrane damage and consequently cell death [15–17], the manifold effects that different interfaces have on amyloid aggregation

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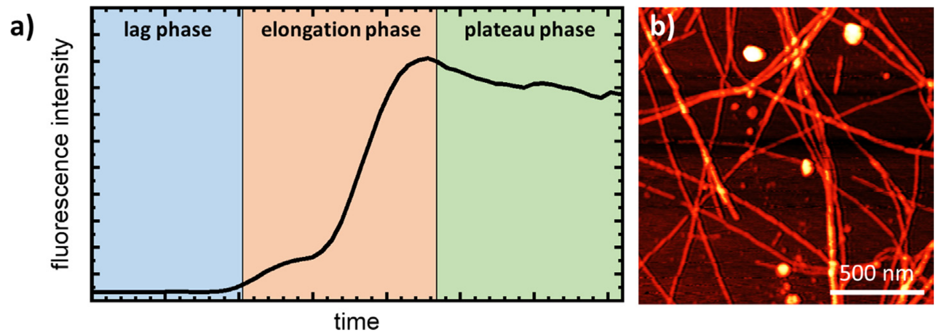

Fig. 1. Aggregation of the type 2 diabetes-related peptide hIAPP(1–37) in bulk solution monitored by thioflavin-T (ThT) fluorimetry (a) and atomic force microscopy (AFM) image of the resulting fibrillar aggregates (b).

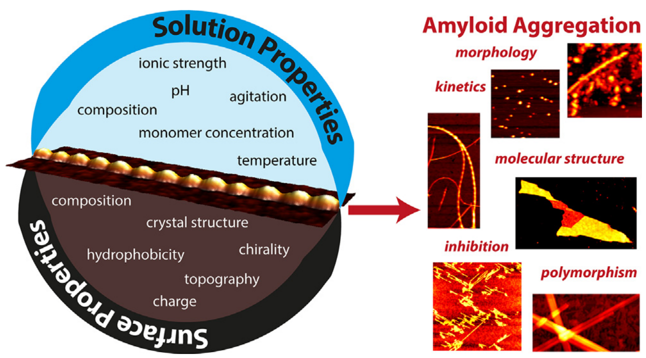

Fig. 2. Left: Overview of solution and surface properties known to affect amyloid aggregation. Right: Amyloid aggregation parameters affected by bulk and solution properties. The AFM images on the right show aggregates of the peptide fragment hIAPP(20–29) obtained under different conditions in contact with various hydrophilic and hydrophobic surfaces.

are still not fully understood [18]. In this context, the physicochemical surface properties such as hydrophobicity, charge, and surface chemistry play important roles since they directly affect the interaction with the monomers. Small variations of these surface properties may thus influence the conformation and mobility of the adsorbed monomers and therefore decide over promotion or inhibition of amyloid aggregation (see Fig. 2). Such surface-specific effects are becoming more and more important due to the emerging use of organic and inorganic nanoparticles in theragnostics and drug delivery [19–21], which may interact with proteins and peptides and thereby modulate their aggregation [22]. Therefore, the interaction of various amyloidogenic peptides and proteins with different organic and inorganic model surfaces and nanoparticles has become the focus of intense research efforts.

Studies that investigate amyloid aggregation at model surfaces typically face two challenges. First, many of the analytical techniques that have been established for characterizing amyloid aggregation in bulk solution cannot or only to a rather limited extent be applied to surfaces. This in particular concerns the widely used thioflavin T (ThT) fluorescence assay [23–25]. Second, controlling and especially varying surface properties in a well-defined way is more difficult than for solution conditions, as it requires sophisticated fabrication and modification techniques. Furthermore, surface properties are often tightly correlated with each other so that it may be virtually impossible to adjust one parameter without affecting another. In the following, we will thus first discuss the pros and cons of a selection of experimental techniques that can be employed in the study of amyloid aggregation at solid-liquid interfaces, before introducing various model surfaces that so far have been utilized in such studies. Then, we will summarize the current state of knowledge regarding amyloid formation at solid-liquid interfaces with a particular emphasis on still open questions. Finally, future perspectives and experimental challenges will be addressed. Note that the remainder of this perspective article will focus on solid surfaces. Phospholipid bilayers have mostly been excluded from the discussion because their interactions with various amyloidogenic proteins and peptides have been studied and reviewed extensively, see for instance references [15–17,26–37].

2. Experimental techniques for studying amyloid aggregation at solid-liquid interfaces

2.1. Fluorescence photometry and microscopy

The most common approach to monitor amyloid aggregation in bulk solution exploits the specific increase of the fluorescence of the dye ThT upon selective binding to β-sheet-containing protein and peptide aggregates [23–25]. Intrinsically a bulk technique typically conducted in a microplate reader, this approach can also be utilized for investigating amyloid aggregation in the presence of solid surfaces, as long as these surfaces are introduced in the form of colloidal particles. Many studies have used this approach quite successfully for studying the effect of various polymeric and inorganic nanoparticles on amyloid aggregation [38–51]. This assay is also frequently used in conjunction with metal nanoparticles, such as gold and silver nanoparticles displaying different surface functionalizations [42,43,52–58]. In this case, however, fluorescence quenching may occur and make the analysis of the ThT fluorescence signal more difficult [59].

The Goto group has combined this established bulk ThT fluorescence assay with total internal reflection fluorescence (TIRF) microscopy [60–64] which enabled them to follow the growth of individual amyloid fibrils at the surfaces of the employed glass slides *in situ* and in real-time. This approach was more recently further advanced toward two-color fluorescence imaging by employing fluorescently labeled islet amyloid polypeptide (IAPP) monomers and non-labelled, preassembled fibril seeds in combination with ThT in order to elucidate the role of lateral monomer binding to the fibril surfaces [63].

The TIRF microscopy approach suffers from a rather low resolution of only a few hundred nanometers, which prevents the visualization of oligomeric amyloid aggregates. Super-resolution fluorescence microscopy techniques such as STORM or PALM on the other hand can overcome this resolution limit [65–67]. A more severe limitation of many fluorescence microscopy techniques with regard to the study of surface effects is their frequent requirement of transparent substrates. This reduces the variety of surface materials that can be investigated in such experiments to a handful of transparent substrates such as silica and alumina that may additionally be coated with thin organic films.

2.2. Atomic force microscopy

The most popular microscopic technique for imaging amyloid aggregates at various surfaces, both *ex situ* and *in situ*, is probably atomic force microscopy (AFM). The advantages of AFM in these kinds of studies are manifold: It provides a lateral resolution down to the few-nanometer range and thus enables the identification and classification of amyloid oligomers, it can be operated under dry and liquid conditions, including physiological buffer solutions, and imaging can be performed on almost any sufficiently smooth substrate surface, organic and inorganic, conducting and insulating, opaque and transparent. Since AFM relies on the mechanical interaction of a scanned tip with the sample surface, it can furthermore provide mechanical information of the surface. Using high-resolution force mapping, different amyloid aggregates can be imaged while simultaneously measuring their elastic moduli with nanometer resolution [68–70].

AFM also allows for the continuous imaging of the surface, thus providing real-time *in-situ* information of adsorption, desorption, diffusion, assembly, and aggregation processes. While AFM is traditionally considered a rather slow technique in which recording a single image takes several minutes, recent advances in high-speed AFM enable the imaging of biomolecular dynamics at scan rates below 1 s per frame [71]. A few studies so far have employed high-speed AFM to successfully monitor the growth of amyloid fibrils [72–74] and even the structural dynamics of various oligomeric species [74–76] in real-time. However, care must be taken when analyzing and interpreting high-speed AFM results due to the possible occurrence of artifacts stemming from disruptive interactions between the biomolecules and the rapidly scanned AFM tip [77].

The application of AFM to study amyloid aggregation at surfaces typically faces two limitations. First, the maximum image size that can be recorded is determined by the range of the employed scanner and typically limited to some 10 μm. Second, in order to reliably identify small oligomeric structures, the substrate surface has to be rather flat. Both of these limitations become even more severe in high-speed AFM, since high scan rates usually require even smaller scan sizes and smoother substrates. This is why all corresponding high-speed AFM studies so far have utilized atomically flat mica substrates and scan sizes well below $3 \times 3\ \mu m^2$ [72–76].

2.3. Surface plasmon resonance

Surface plasmon resonance (SPR) is a widely used bioanalytical assay for the detection of biomolecular species and the quantification of receptor-ligand affinities [78,79]. It is based on the resonant induction of surface plasmons in thin metal (typically gold) films via incident light. Surface plasmon formation depends strongly on the refractive index of the medium in the direct vicinity of the metal surface, slight variations of which – for instance due to biomolecular adsorption – can thus be detected by a change in the intensity of the reflected light. This change is a measure of adsorbed mass. In this way, biomolecular adsorption and desorption can be monitored *in situ*.

SPR has been employed quite extensively for the detection of toxic amyloid species via specifically binding ligands such as antibodies [80–83] and to study the growth of fibrils from surface-tethered monomers and fibrils [84–87]. While the application of SPR in the study of amyloid aggregation at different surfaces has already been demonstrated [88], such studies are restricted to very thin organic or inorganic films such as self-assembled monolayers that can be deposited on the surfaces of the metal-coated prisms. Since only changes in the integral optical properties related to the adsorbed mass are detected, however, adsorption of monomers, oligomers, and pre-assembled fibrils cannot be distinguished. Furthermore, aggregation and fibrillization of adsorbed monomers and oligomers cannot be detected at all. Therefore, SPR studies of amyloid aggregation at solid-liquid interfaces typically require the application of complementary techniques such as AFM in order to correlate adsorption and aggregation [88].

2.4. Quartz crystal microbalance

Quartz crystal microbalance (QCM) is a technique that provides largely similar information as SPR in that sense that it enables the *in-situ* monitoring of adsorption and desorption processes [89]. This is achieved by monitoring the resonance frequency of the shear oscillation of a piezoelectric quartz crystal over time, which depends on the overall mass of the crystal. Changes in adsorbed mass thus result in a shift of the resonance frequency. Compared to SPR, QCM has two major advantages. First, the electrode surfaces of the QCM sensors can be coated with comparatively thick films (several hundred nanometers in thickness) of almost arbitrary materials, such as metals, alloys, oxides, semiconductors, polymers, and molecular layers, without affecting the sensitivity of the measurement. Second, in addition to the frequency shift, the dissipated energy of the oscillating sensor surface can be measured as well. This approach is typically called quartz crystal microbalance with dissipation monitoring (QCM-D). Since the change in dissipated energy is a measure of the viscoelastic properties of the adsorbed film, QCM-D has become a very powerful tool for studying the adsorption of proteins and other biomolecules which tend to form adsorbate films of rather high viscoelasticity [89].

Several studies have so far employed QCM-D to investigate amyloid aggregation at interfaces. While some of them have again focused on quantifying the growth rates of surface-tethered fibrils [90–92], several others employed QCM-D to investigate amyloid aggregation in contact with various model surfaces, in particular supported lipid bilayers and SAMs but also metal oxides [93–101]. The latter studies in particular benefit from monitoring the change in dissipation which can be used to distinguish between the adsorption of different amyloid species [95] and characterize fibril growth and conformation in more detail [93,94,100]. Unfortunately, fibril formation may result in rather complex sensor responses whose analyses often require the application of sophisticated models [93,100]. Therefore, *in-situ* QCM-D measurements of amyloid aggregation are often complemented with *ex-situ* AFM or fluorescence microcopy [93,95–101]. Recently, a combined QCM-TIRF microscope has been developed, which enabled the simultaneous measurement of deposited peptide mass on the QCM sensor surface and visualization of fibril growth by TIRF microscopy [102].

2.5. Circular dichroism and infrared spectroscopy

All the techniques discussed so far suffer from an inability to provide molecular information of the aggregate structures. In bulk solution, such information is typically obtained either *in situ* from circular dichroism (CD) spectroscopy or *ex situ* by attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy [103,104], both of which are used to quantify secondary structure components and in particular intermolecular β sheets. CD spectroscopy has also been used successfully for studying amyloid aggregation in the presence of various nanoparticles in bulk solution [49,105–107]. When it comes to thin adsorbate films on flat surfaces, however, the comparatively low sensitivities of both CD and ATR-FTIR spectroscopy often become a problem. While most CD spectrometers are suited only for bulk measurements, ATR-FTIR is intrinsically an interface-sensitive technique [108]. In contrast to TIRF or SPR, however, the evanescent IR wave typically penetrates several hundred nanometers up to few microns into the optically less dense medium across the crystal surface, so that only a rather small fraction of the overall signal actually stems from the few nanometers thick adsorbate film. However, the measurement of difference spectra allows for the separation of bulk and interface related spectral responses. Moreover, the use of polarized light enables the analysis of the orientation of adsorbed molecules with respect to the surface plane [109,110]. The sensitivity of such studies for the interface layers depends on various parameters such as the number of internal reflections, the refractive index of the optical crystal, the angle of incidence and the presence of additional metal films applied on the crystal [111]. For details, we refer to excellent reviews on ATR-FTIR spectroscopy and its application for studying protein adsorption [110,112]. Furthermore, it has been reported that the plane contact of a material with even higher refractive index leads to an enhancement of the signal of a layer sandwiched between this substrate and the ATR-crystal [113,114]. Generally, a strong advantage of ATR spectroscopy lies in the possibility to analyse adsorption processes from extended electrolyte phases. Finally, the interface sensitivity of FTIR-ATR spectroscopy can be further improved based on the preparation of coinage metal island film on the surface of the optical crystal [115]. In such cases the surface enhanced infrared spectroscopy (FTIR-SEIRAS) can be performed in the ATR configuration [111,116,117].

In principal, an even more interface-sensitive technique than ATR-FTIR is polarization modulation infrared reflection absorption spectroscopy (PM-IRRAS). Here, a linearly polarized IR beam is reflected from a metallic surface. Parasitic absorptions from the surrounding atmosphere or electrolyte can be canceled by the analysis of the differences in the reflectance of the parallel and the perpendicular polarized light. Only the parallel component contains information of the adsorbate while the isotropic bulk phase contributes equally to both polarizations. The interface enhancement is based on the summation of the field vector perpendicular to the substrate surface and the increase in the analyzed interface area due to the grazing incidence [118]. While this technique typically yields high quality IR spectra even at very low surface coverage [100,101], it requires a highly reflecting metallic substrate that can be coated with ultra-thin films of the desired composition, thereby again limiting the range of materials to be investigated. Furthermore, polarization modulation results in the peculiarity that only dipole orientations perpendicular to the surface contribute to the overall spectrum, which may lead to discrepancies when complemented with other methods [101]. However, as for ATR spectroscopy the dipole selection rule might be useful for the analysis of molecular orientations [119]. It should be noted that for the analysis of electrolyte/solid interfaces a quite small electrolyte gap between the optical window and the reflecting substrate is required to minimize the undesired absorbance of light in the bulk electrolyte.

Both ATR-FTIR and PM-IRRAS provide ensemble spectra that do not account for the possibility of variations in the molecular structures of different aggregates species [7,9]. Nearfield techniques that combine IR spectroscopy with scanning probe microscopy such as AFM-IR [120] and Nano-FTIR [121], on the other hand, allow for recording localized IR spectra of single protein complexes with a lateral resolution of few ten nanometers. Thereby, IR spectra of single amyloid fibrils can be recorded and correlated with their morphology [122–125].

2.6. Raman spectroscopy

While Raman spectroscopy provides essentially the same secondary structure information as IR spectroscopy, it is less sensitive for thin films and thus less suited for the analysis of protein aggregates at surfaces. Surface-enhanced Raman scattering (SERS), on the other hand, utilizes the electromagnetic field enhancement in the close vicinity of metallic surfaces and nanostructures to obtain higher Raman intensities and thus reach single-molecule sensitivity [126]. Although SERS-detection of amyloid aggregates has been demonstrated using nanorough gold surfaces [127], the application of various types of nanoparticles is more common as higher Raman signal intensities are typically achieved in this way [128–133].

A general limitation of the SERS approach is the strong dependence of the electromagnetic field enhancement on the distance from the surface. In general, the analyte molecules need to be located not more than few nanometers from the surface. This implies that although the SERS-active surfaces can be coated, the applied coatings have to be extremely thin with thicknesses in the range 1–2 nm. On the other hand, SERS-active nanoparticles may also be deposited on top of a molecular adsorbate film formed on an arbitrary surface for *ex-situ* analysis [134].

Similar to the case of Nano-FTIR discussed above, the SERS approach can be combined with AFM to obtain localized Raman spectra. This technique is called tip-enhanced Raman spectroscopy (TERS) and utilizes a gold- or silver-coated AFM tip that is scanned across the surface-immobilized analyte as a SERS-active probe [135]. This allows the correlation of aggregate morphology and secondary structure components [136,137]. Since TERS can achieve much higher lateral and vertical resolution than AFM-IR and Nano-FTIR [138], it can also be used to map the amino acid residue composition of the fibril surface [139–141]. However, obtaining high-quality spectra by TERS is experimentally more demanding than by IR nanospectroscopy as it is more sensitive toward tip properties and requires a very precise optical alignment.

3. Types of model surfaces: Tuning physicochemical surface properties

3.1. Mica

Muscovite mica is a common substrate for the investigation of biomolecules by AFM [142]. Due to its layered crystal structure, it can be easily cleaved to generate a clean surface with large, atomically flat terraces. Furthermore, it has a strongly negative net charge in aqueous solutions, which enables efficient adsorption of charged biomolecules such as proteins and DNA. Consequently, mica is routinely used as a substrate in the AFM-based investigation of amyloid aggregation, where its flat surface enables not only the identification and distinction of oligomers, protofibrils, and mature fibrils [8,68,69,75,143–155], but also the application of high-speed AFM [72,73]. There is also the possibility to modify the mica surfaces, for instance by the application of organosilane monolayers [156] or ion beam treatment [157], in order to fabricate flat model surfaces with variable physicochemical properties to study their effects on amyloid aggregation in detail [145,147].

3.2. Highly oriented pyrolytic graphite

Similar to mica, highly oriented pyrolytic graphite (HOPG) has a layered crystal structure and can thus be cleaved, resulting in an atomically flat surface. It is thus another well-established substrate for AFM investigations [99,155,158–160]. However, in contrast to the negatively charged, hydrophilic aluminosilicate surface of mica, HOPG is hydrophobic. It is therefore sometimes used along mica as a hydrophobic model surface of similar flatness [146,148]. Care should be taken, however, when comparing these two model surfaces, as their chemical properties differ significantly, so that observed differences in the adsorption or aggregation behavior of amyloidogenic proteins or peptides may not exclusively be caused by the differences in hydrophobicity. Furthermore, it is sometimes observed that proteins have a higher affinity for the step edges of HOPG surfaces than for the atomically flat terraces [161], which may locally promote amyloid aggregation at these surface sites [159].

3.3. Silica

Silica is another comparatively flat, negatively charged model surface frequently employed in in the AFM investigation of biomolecules [142]. In aqueous solution, the charge of the silica surface depends on the surface density of silanol groups and thus on the pH of the electrolyte. The point of zero charge (pzc) of silica is typically around pH 3, implicating a negative net charge under physiological pH. However, even under strongly alkaline conditions, the overall charge density of silica is significantly lower than that of mica [162,163].

Two sub-types of silica surfaces need to be distinguished: oxidized silicon and glass. Silicon wafers represent an extremely well defined substrate with a rather flat surface that typically has a root-meansquare (rms) roughness of a few Angstrom. Furthermore, due to its tremendous importance in semiconductor industry, oxidization of silicon is a well-studied process, for which highly reproducible protocols are available that yield surfaces with virtually identical properties [164]. Glass, on the other hand, is naturally been used in studies employing optical techniques such as TIRF microscopy [60,61,63,64]. Unfortunately, commercial glass substrates are much less defined than silicon wafers and can have very different chemical compositions and surface topographies, which reduces comparability. Even worse, it is known for several decades, that the bulk and surface properties and in particular the surface energy of SiO_2 depend strongly on the nature and history of the specimen [165–167], which renders uncoated glass a less than optimal substrate for investigating adsorption phenomena. Silanization of the glass surfaces, on the other hand, enables the fabrication of chemically better defined surfaces for TIRF microscopy investigations [62].

3.4. Polymers

Polymer surfaces have long been employed to study the influence of different surface properties on amyloid aggregation [38–41,44,49,105–107,155,168–174]. Often, polymeric micro- or nanoparticles were simply added to a bulk sample of amyloidogenic proteins or peptides to monitor their effects by ThT fluorescence and other bulk techniques. However, also flat polymer surfaces can be fabricated by coating a suitable substrate with a thin polymer film, for instance by spin coating [171,172]. Due to the multitude of different polymer species and building blocks that are commercially available, polymer surfaces with very different chemical composition, hydrogen bonding capacity, charge, and hydrophobicity can be synthesized. This great versatility, however, often faces the challenge of adjusting one of these parameters without affecting the others. Such parameter crosstalk often makes it difficult to rationalize observed differences in adsorption or aggregation behavior. Furthermore, controlling the surface topography of polymer films and particles may be rather difficult, so that different surfaces may exhibit very different roughness and texture.

3.5. Metals and metal oxides

Amyloid aggregation at metal and metal oxide surfaces has mostly been investigated for nanoparticle systems, since gold, silver, Fe_2O_3, and other metal-based nanoparticles are of high medical relevance. In most cases, however, those nanoparticles were modified with capping ligands to prevent nanoparticle aggregation during the experiments, so that the metal or metal oxide surface of the nanoparticle did not come into direct contact with the proteins or peptides under investigation [42,43,52,54,56–58,175–177]. Flat metal surfaces, on the other hand, are usually only used when the employed technique demands for it, for instance in the form of the gold electrodes of QCM-D sensors or the gold films on SPR waveguides. The majority of such studies, however, used gold surfaces modified with thin coatings such as self-assembled thiol monolayers presenting selected chemical groups [84,88,95,100,101,127,178]. Amyloid aggregation at flat metal oxide surfaces has so far only rarely been investigated, and mostly with regard to medically relevant implant materials such as Ta_2O_5 [93,99]. Despite that, metal oxide surfaces represent rather interesting model surfaces as they are typically fairly well characterized with regard to surface structure and chemical composition, have widely differing pzc values, can be prepared comparatively straightforward on almost arbitrary substrates using atomic layer deposition [179], and thus allow for the application of numerous complementary surface analytical techniques.

3.6. Self-assembled monolayers

As already briefly mentioned above, self-assembled monolayers (SAMs) are powerful tools for modifying the chemical properties of solid surfaces in a very precise way [180,181]. Mostly, they are applied on gold and oxide surfaces using organothiol and organosilane functionalities, respectively, and may display various functional groups. Because of the versatility of SAM formation, this approach is fully compatible with almost all surface sensitive techniques, including TIRF microscopy, AFM, SPR, QCM-D, and PM-IRRAS [62,88,100,101,178,182]. Polar, negatively charged, positively charged, and hydrophobic surfaces can be fabricated in this way by SAMs displaying OH, COOH, NH_2, and CH_3 groups, respectively. SAMs thus enable very detailed studies of surface effects on amyloid aggregation by employing both molecularly well-defined surfaces with tunable properties and various complementary analytical techniques that allow access to adsorption and aggregation kinetics, aggregate morphology, and molecular aggregate structure.

3.7. Nanoparticles

The interaction of nanoparticles with proteins and peptides has become a topic of intense research in the last years [183]. Just like flat surfaces, nanoparticles can have a tremendous effect on amyloid aggregation, with the surface properties discussed so far, *i.e.*, chemical composition, polarity, charge, hydrophobicity, and hydrogen bonding capacity, being important parameters [184]. Since they also allow for the application of bulk techniques such as ThT fluorimetry and CD spectroscopy, colloidal nanoparticles have been utilized early on for investigating amyloid aggregation at interfaces [105,106]. In contrast to flat surfaces, however, amyloid-nanoparticle interactions are significantly more complex, since nanoparticles carry additional properties that may affect aggregation kinetics and aggregate structure, namely nanoparticle size and shape. These issues have recently attracted a lot of attention, in particular since it was recognized that nanoparticle size can decide over amyloid inhibition and promotion [44,57,132], rendering biomedical nanoparticles both potential threads and panaceas.

Another important factor that distinguishes most nanoparticles from flat surfaces is their surface chemistry. In order to avoid aggregation and segregation of the nanoparticles in physiological salt concentrations, their surfaces are usually terminated with stabilizing, hydrophilic capping ligands [185,186]. Since different synthesis routes often utilize different capping ligands [187], the physicochemical surface properties of a given nanoparticle system, most importantly chemical composition, charge, and hydrogen bonding capacity, and thus its interaction with biological systems may vary drastically [188–190]. In order to obtain meaningful data and correlations, the surface properties of the nanoparticles therefore need to be thoroughly characterized. Unfortunately, however, it is often very difficult to control ligand properties and surface density independently of nanoparticle size and shape. Due to this large variety of relevant parameters and the strong influence of experimental and environmental conditions, no coherent picture of the influence of nanoparticles on amyloid aggregation could be established yet (see below).

4. Amyloid aggregation at solid surfaces: A barely understood phenomenon

4.1. Epitaxial amyloid growth on single-crystalline surfaces

One factor that is frequently neglected in the investigation of protein-surface interactions is the crystallinity and crystal structure of the surface. However, already in 1999, Kowalewski and Holtzman have shown that the peptide Aβ(1–42) produces sheet-like amyloid-like structures on the HOPG surface that follow certain crystallographic directions [148]. This effect, in analogy to semiconductor and metal epitaxy termed epitaxial amyloid growth, was subsequently observed for a number of biological and artificial peptides and proteins, both on

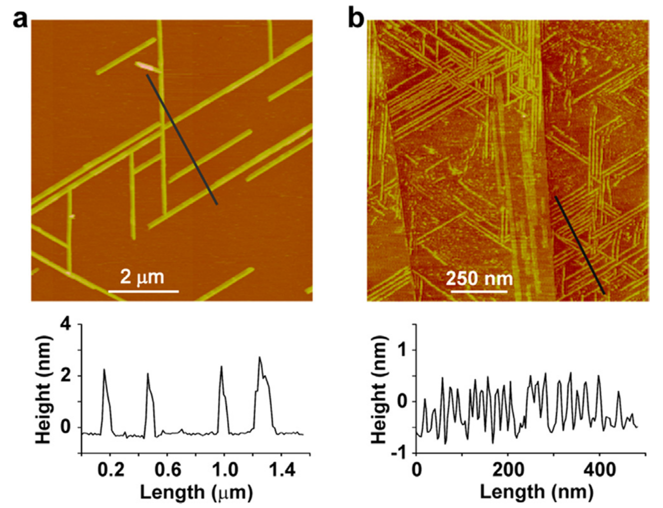

Fig. 3. AFM images of P4 peptides epitaxially grown on a mica (a) and a HOPG surface (b), respectively, with corresponding line scans take along the black lines. Preferential alignment of the sheet-like amyloid structures along certain crystallographic directions of the surfaces is observed. Reprinted with permission from [193]. Copyright (2012) American Chemical Society.

HOPG and on mica surfaces (see Fig. 3) [145,146,151,191–197]. On HOPG, attractive hydrophobic interactions between hydrophobic amino acid residues and the hydrophobic HOPG surface result in a comparatively flat configuration of the assembled amyloid sheets [192,193,196]. On the negatively charged mica surface, epitaxial growth results from electrostatic interactions that may involve both charged amino acid residues and the charged peptide termini [145,191–194,196]. The latter phenomenon can be controlled to some extent by tuning the strength of the electrostatic interactions through chemical modification of the peptide or variation of the buffer's salt concentration [191,194], which makes this epitaxial amyloid growth an interesting technique for nanomaterials and nanopatterning applications. However, further studies are needed to determine whether such epitaxial amyloid growth may also occur on technologically more relevant surfaces.

4.2. The influence of hydrophobic and electrostatic interactions

The example of epitaxial amyloid growth demonstrates the importance of hydrophobic and electrostatic interactions between the surface and the protein or peptide. Since these interactions strongly depend on the sequence and structure of the peptide/protein, the properties of the surface, and the environmental conditions such as ionic strength and pH of the buffer, however, it is very difficult to derive a general understanding of the molecular mechanisms involved.

For instance, aggregation of the type 2 diabetes-associated human islet amyloid polypeptide hIAPP(1–37) on ion beam-modified mica surfaces in water was found to be drastically affected by small increases in hydrophobicity [147]. Increasing the contact angle of the surface from 23° to 38° resulted in a measurable delay in fibrillization and in fibrils of smaller diameter. Further increases in hydrophobicity resulted in a continuous delay in fibrillization, until at contact angles > 75°, a transition in the pathway of aggregation was observed, leading to the disappearance of fibrillary aggregates. At fully hydrophobic surfaces, multilayers of comparatively large oligomers were observed. On the same type of surface, fibrillization of the amyloidogenic hIAPP(1–37) fragment hIAPP(20–29) was found to be delayed as well [145]. In contrast to the full-length peptide, however, fibrils, larger amorphous aggregates, as well as oligomers were found to coexist on this hydrophobic surface. Hydrophobic, CH_3-terminated SAM surfaces also delayed fibrillization of hIAPP(1–37) in PBS buffer compared to OH, COOH, and NH_2 SAMs, but did not completely inhibit fibril formation (see Fig. 4, upper row) [100]. The same experiments conducted with hIAPP(20–29) at a concentration that did not result in fibrillization in bulk solution, however, yielded rather surprising results. Here, the hydrophobic CH_3 SAM induced the formation of large fibrillar aggregates, while the other surfaces barely showed any aggregates at all (Fig. 4, lower row) [101].

Similarly confusing observations have been made with regard to the relevance of electrostatic interactions. Jeworrek *et al.* observed that adsorption of the positively charged hIAPP(1–37) to negatively charged polymer surfaces was completely abolished by addition of 100 mM NaCl [168]. Hajiraissi *et al.*, on the other hand, measured strong hIAPP (1–37) adsorption to negatively charged COOH-terminated SAMs in physiological PBS buffer with 137 mM NaCl, which was accompanied by strong fibrillization (see Fig. 4, upper row) [100]. Even more surprising, the same authors observed strong hIAPP(1–37) adsorption and fibrillization on positively charged NH_2-terminated SAMs, despite hIAPP(1–37) carrying only positive charges, but no negative ones (Fig. 4, upper row). This was explained by hydrogen bond formation between peptide and SAM, which may overcome electrostatic repulsion [100]. Similar observations were made for hIAPP(20–29), which carries no charged amino acids, so that electrostatic interactions with surfaces have to be mediated via the termini. Here, it was for instance observed that a peptide with amidated C terminus, which thus carries only a positive charge at the N terminus, shows *lower* adsorption to negatively charged COOH SAMs than the same peptide with acetylated N terminus and native, negatively charged C terminus [101]. The native peptide with a positive and a negative charge at the N and the C terminus, respectively, showed the highest adsorption, together with the completely uncharged peptide [101].

These examples show how difficult it is to dissect the relative

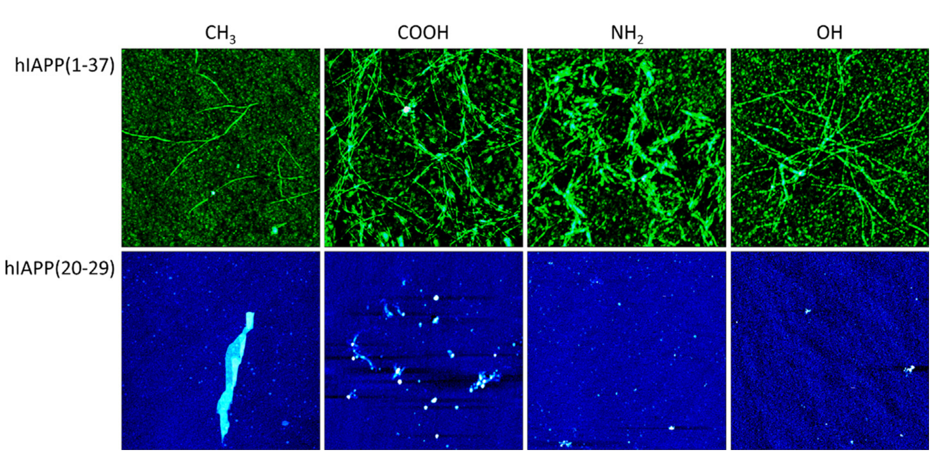

Fig. 4. AFM images of hIAPP(1–37) (upper row) and hIAPP(20–29) (lower row) incubated in PBS buffer at 37 °C on different SAMs for ~400 min and ~800 min, respectively. Images have a size of $4 \times 4\,\mu m^2$ (upper row) and $10 \times 10\,\mu m^2$ (lower row), respectively. Upper row adapted with permission from [100]. Copyright (2018) American Chemical Society. Lower row adapted with permission from [101] (https://pubs.acs.org/doi/10.1021/acsomega.8b03028). Further permissions related to the material excerpted should be directed to the American Chemical Society.

contributions of the different molecular interactions to amyloid aggregation at liquid-solid interfaces. In order to truly understand the molecular mechanisms involved in amyloid aggregation at interfaces, systematic investigations under identical experimental conditions are urgently needed that use various proteins and (model) peptides, as well as well-characterized, molecularly defined surfaces. Further help might come from molecular dynamics simulations as was the case for the field of epitaxial amyloid growth [191–193], provided that simulation and experiment address comparable scenarios and conditions.

4.3. The importance of surface mobility

Despite the difficulties in understanding the role of individual interactions in amyloid aggregation at surfaces, some general insights into the effects of interaction strengths have been obtained. It is rather accepted nowadays, that fibrillization at interfaces requires weak attractive interactions between peptide or protein and solid surface [100,145,152–154,172]. This will lead to the adsorption of monomers, which maintain a high degree of lateral mobility at the interface. The latter enables them to diffuse along the interface until they encounter and attach to another monomer, an oligomer, or a fibril, resulting in aggregate growth. This monomer attachment of course involves conformational rearrangements of the monomer, which again require a certain degree of mobility. If the strength of the attractive interaction is too weak, adsorption and thus aggregation will be negligible. If the interaction is too strong, on the other hand, lateral mobility will be insufficient to allow for diffusional movement and conformational alterations. At intermediate interaction strength, however, the adsorbate film can be understood as a two-dimensional fluid with an increased concentration compared to the bulk phase. This locally increased concentration in combination with the high two-dimensional mobility of the monomers may then result in strong fibrillization even under conditions that do not favor aggregation in bulk [154,172]. In this intermediate, weak interaction regime, however, small variations in the interaction strength, the type of interaction, or the relative contributions of different interaction types may lead to drastic and hard to predict changes in aggregation kinetics and aggregate structure and morphology [147].

Due to the high importance of monomer mobility and diffusion, lateral variations in surface properties may modulate amyloid aggregation. Recently, Shezad *et al.* investigated the effect of surface roughness on the aggregation of Aβ(1–42) on polymer surfaces [171]. They found that fibrillization propensity scales inversely with surface roughness, with rougher surfaces retarding or even inhibiting amyloid aggregation by hindering interface diffusion (see Fig. 5). Surface topography thus seems to play an important role in amyloid aggregation at interfaces and deserves further in-depth investigation. In particular, surface topography can be described by various statistical quantities such as rms roughness, roughness exponent, correlation length, skewness, kurtosis, fractal dimension, etc [198]. All of these more or less independent parameters may affect amyloid aggregation in one way or another and thus need to be studied individually or combinatorically using well-defined model surfaces with varying, yet fully characterized surface topography.

4.4. The effect of chirality

Mature amyloid fibrils often are chiral [199], as are the lipids of biological membranes they interact with [200]. It was also observed that the chirality of amyloid inhibitors has a strong effect on their potency [201]. Consequently, a few recent studies have investigated the aggregation of amyloidogenic peptides at surfaces modified with different enantiomers of chiral molecules [202–206]. Using graphene oxide surfaces modified with R-cysteine, S-cysteine, and several derivatives thereof, Qing *et al.* showed that surface chirality has a large effect on Aβ(1–40) aggregation [203]. The authors in particular

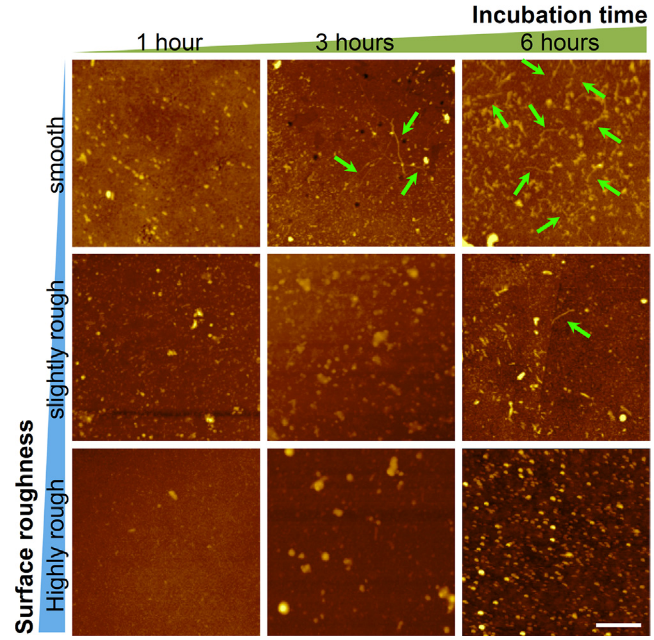

Fig. 5. AFM images of Aβ(1–42) aggregates obtained after different incubation times on polystyrene surfaces with different roughness. Green arrows point to fibrillar aggregates. The scale bar is 400 nm for all panels. Reproduced with permission from [171]. Copyright (2016) American Chemical Society. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

observed that the R-cysteine-modified surface suppressed Aβ(1–40) adsorption and aggregation, whereas the S-cysteine-modified surface was promoting amyloid formation. They furthermore showed that this chirality effect is indeed a surface effect, which could be abolished by inserting a molecular spacer between enantiomer and surface. Nevertheless, similar observations have been made also on R- and S-cysteine-modified SiO_2 surfaces [206]. On gold surfaces modified with N-isobutyryl-L-cysteine (L-NIBC), Aβ(1–40) formed ring-like aggregates whereas corresponding aggregates on N-isobutyryl-D-cysteine (D-NIBC) surfaces exhibited a rod-like morphology [205]. Although both aggregate morphologies had similar mechanical properties and secondary structure characteristics, they differed substantially from those of mature Aβ(1–40) fibrils formed in bulk solution. When immobilized on gold nanoparticles, both enantiomers were able to inhibit hIAPP fibrillization, with L-NIBC showing higher potency [204]. Du *et al.* furthermore showed that cross-seeding of Aβ(1–40) and insulin on a D-phenylalanine-modified mica surface resulted in cytotoxic fibrillary aggregates, whereas only non-toxic amorphous aggregates were observed on an L-phenylalanine surface [202]. Interestingly, the authors found that the effect of chirality depends also on the nature of the chiral amino acid used for surface functionalization. These peculiar and numerous effects of surface chirality are not fully understood yet and deserve further experimental and theoretical investigations.

4.5. The complicated tale of nanoparticle-amyloid interactions

As already briefly mentioned above, amyloid aggregation in the presence of nanoparticles is a highly complex phenomenon that depends on a multitude of parameters. In addition to the various bulk properties, also nanoparticle concentration, material, capping ligand, charge, hydrophobicity, size, and shape influence aggregation kinetics and aggregate structure in molecule-specific ways. Aβ(1–40) is probably the best studied peptide with regard to the effect of nanoparticles on amyloid aggregation [40,41,48–51,56,57,105–107,177]. Cabaleiro-Lago *et al.* for instance investigated Aβ(1–40) aggregation in the presence of copolymeric nanoparticles with 40 nm diameter and different

NiPAM:BAM ratios [41]. While these hydrophobic nanoparticles were able to increase the lag phase of Aβ(1–40) aggregation, the elongation phase remained largely unaffected. Interestingly, however, the retarding effect of the hydrophobic nanoparticles decreased with increasing hydrophobicity. This was attributed to the higher BAM content in the more hydrophobic nanoparticles, which reduced their ability to form hydrogen bonds. Skaat *et al.* studied the interaction of hydrophobic, FF dipeptide-coated 57 nm nanoparticles with Aβ(1–40) and found a similar retarding effect on the lag phase [49]. In contrast, AA dipeptide-presenting nanoparticles were found to promote Aβ(1–40) aggregation [49].

For negatively charged nanoparticles, the situation is more complex. While 86 nm silica nanoparticles with a zeta potential of −40 mV had no effect on Aβ(1–40) aggregation, fluorinated micelles with 3–5 nm diameter and zeta potentials of about −47 mV efficiently inhibited aggregation [107]. Hydrogenated micelles of same size but lower zeta potential (−20 to −25 mV), however, did not show any inhibitory effect [107]. Yoo *et al.* on the other hand, have demonstrated inhibition of Aβ(1–40) aggregation by thioglycolic acid-capped CdTe quantum dots with a size and zeta potential of 3.5 nm and –31 mV, respectively [51]. This indicates the critical roles of zeta potential and/or surface chemistry. Chan *et al.* compared the effect of quantum dots and gold nanoparticles with different capping ligands on Aβ(1–40) aggregation [177]. They observed that 2.8 nm NAC-capped quantum dots retarded aggregation, whereas 2.2 nm MPA-capped quantum dots did not, despite the similar zeta potentials of –30 and –25 mV, respectively. This again was attributed to different hydrogen bonding abilities of the two capping ligands. Interestingly, slightly larger quantum dots of 3.1 nm size did not inhibit aggregation, irrespective of the capping ligand, which the authors rationalized by the lower zeta potentials of these quantum dots (–37 and –40 mV, respectively). Even larger MPA- and NAC-capped quantum dots 3.5 and 3.8 nm in size, respectively, were however found to promote Aβ(1–40) aggregation, despite their zeta potentials of –30 and –29 mV, respectively, being similar to those of the small quantum dots. In the case of MPA-, NAC-, and citrate-capped gold nanoparticles of 15 nm size, the same authors observed no significant effect on aggregation, even though these nanoparticles had similar zeta potentials as the quantum dots, *i.e.*, –49, –25, and –39 mV, respectively. On the other hand, Liao *et al.* observed that negatively charged carboxyl-capped gold nanoparticles of 30 nm diameter inhibited Aβ(1–40) aggregation while positively charged amine-capped ones had no significant effect [56]. In contrast, Cabaleiro-Lago *et al.*, observed that amino-modified polystyrene nanoparticles are able to both enhance and retard Aβ(1–40) aggregation [40]. The transition between both effects occurred at a certain nanoparticle surface area in solution with lower and higher nanoparticle surface areas resulting in enhanced and retarded fibrillization, respectively, while the nanoparticle size (57, 120, and 180 nm) had no additional effect. Gao *et al.* recently investigated the effect of nanoparticle size on the aggregation of Aβ(1–40) using glutathione-capped gold nanoparticles with diameters ranging from 1.9 to 36.0 nm and observed enhanced aggregation for the larger nanoparticles while the smallest ones let to complete inhibition, despite the nanoparticle surface area was kept constant [57].

These examples demonstrate the high complexity of the amyloid-nanoparticle interaction, which so far has prevented a coherent description of the involved processes. Furthermore, the different experimental conditions used by the different authors seriously limit the comparability of the experimental results, thereby further complicating the identification of the involved molecular mechanisms and their relative importance. Therefore, systematic studies investigating the effects of various different nanoparticles with different sizes and surface functionalizations are urgently needed. Another parameter, which is known to affect nanoparticle interactions with proteins [207–210] but has so far only rarely been studied with regard to amyloid aggregation [216], is nanoparticle shape. Gold nanoparticles are particularly promising candidates for systematic studies of the impact of nanoparticle shape on amyloid aggregation since they can be synthesized in a broad variety of shapes and sizes [211,212], while their surface chemistry can be varied almost independently of their shape by different approaches such as ligand exchange reactions [213] and layer-by-layer deposition techniques [214,215].

5. Conclusion and future challenges

Amyloid aggregation at solid-liquid interfaces represents a medically and technologically important, yet barely understood phenomenon. In order to derive a consistent and complete picture of the involved mechanisms and effects at a molecular level, systematic studies under well-defined and well-controlled conditions are needed that use a number of complementary techniques to assess aggregation kinetics, aggregate morphology, and aggregate molecular structure. This in particular concerns nanoparticles, which come in numerous sizes, shapes, materials, and surface chemistries, and thus represent a particularly challenging system. However, also studies that focus on flat surfaces face various challenges. Most importantly, novel types of model surfaces need to be developed, which allow for the controlled variation of a single parameter such as hydrophobicity or chemical surface composition independently of their other parameters. The employed model surfaces furthermore need to be thoroughly characterized with regard to their physicochemical properties. This in particular concerns surface topography, which, as we now begin to realize, has a strong effect on amyloid aggregation, but is typically rather difficult to control. Frequently employed measures such as the rms roughness do not provide a complete description of surface topography as they neglect many important details. Rather, complete statistical characterization of surface topography is required to enable unambiguous identification of clear correlations between amyloid aggregation and the numerous morphological parameters such as fractal dimension or correlation length. The so obtained understanding of the molecular mechanisms that govern amyloid aggregation at interfaces will not only provide deeper insights into the medically relevant interaction of amyloid aggregates with cell membranes, but may also enable the synthesis of novel nanoparticle-based therapies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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